

Day 9 Notes – IP Addressing Concepts

1. Introduction to Networking

Networking means communication between:

- Servers
- Computers
- Routers
- Cloud systems

Every device needs:

- Unique identification

That identification is:

IP Address

2. What is an IP Address?

IP = Internet Protocol

Example:

192.168.1.10

Purpose:

- Device identification
- Communication between systems

Like:

- House address for humans
 - IP address for computers
-

3. IPv4 Address Structure

IPv4 contains:

32 bits

Divided into:

4 octets

Example:

192.168.1.10

Each octet:

8 bits

Total:

$8 + 8 + 8 + 8 = 32$ bits

4. Binary Basics

Computers understand only:

0 and 1

This is called:

Binary

5. Binary Number Table

Decimal	Binary
128	10000000
64	01000000
32	00100000
16	00010000
8	00001000
4	00000100
2	00000010
1	00000001

6. Decimal to Binary Example

Convert:

192

Binary:

128 + 64
= 192

11000000

7. Full IPv4 Binary Conversion

IP:

192.168.1.10

Binary:

192 = 11000000
168 = 10101000
1 = 00000001
10 = 00001010

Complete:

11000000.10101000.00000001.00001010

8. IPv4 Classes

Old networking uses:

- Class A
 - Class B
 - Class C
-

9. Class A IP Address

Range

1.0.0.0 to 126.255.255.255

Default Mask

255.0.0.0

CIDR

/8

Binary

11111111.00000000.00000000.00000000

Example

IP:

10.0.0.1

Network part:

10

Host part:

0.0.1

Total Networks

126

Hosts Per Network

16 million+

10. Class B IP Address

Range

128.0.0.0 to 191.255.255.255

Default Mask

255.255.0.0

CIDR

/16

Binary

11111111.11111111.00000000.00000000

Example

IP:

172.16.0.1

Network:

172.16

Host:

0.1

Hosts

65534

11. Class C IP Address

Range

192.0.0.0 to 223.255.255.255

Default Mask

255.255.255.0

CIDR

/24

Example

IP:

192.168.1.10

Network:

192.168.1

Host:

10

Hosts

254

12. Class Comparison Table

Class	Range	Default Mask	CIDR	Hosts
A	1-126	255.0.0.0	/8	16M
B	128-191	255.255.0.0	/16	65K
C	192-223	255.255.255.0	/24	254

13. Private IP Address Ranges

Reserved for internal usage.

Class	Range
A	10.0.0.0/8
B	172.16.0.0 - 172.31.255.255
C	192.168.0.0/16

Used in:

- AWS
 - Offices
 - Data centers
 - Home networks
-

14. Public IP Addresses

Accessible from internet.

Example:

52.x.x.x

AWS public IPs:

- Elastic IP
 - Auto-assigned IP
-

15. Subnet Mask

Subnet mask separates:

- Network portion
- Host portion

Example:

255.255.255.0

Binary:

11111111.11111111.11111111.00000000

16. CIDR Notation

CIDR:

192.168.1.0/24

Meaning:

24 bits network
8 bits host

17. Host Formula

Formula:

$2^{(32-CIDR)}$

Example:

/24

Calculation:

$2^{(32-24)}$
 $= 2^8$
 $= 256$

Usable:

254

18. Common CIDR Values

CIDR	Total	IPs Usable
/30	4	2
/29	8	6
/28	16	14
/27	32	30
/26	64	62
/25	128	126
/24	256	254
/16	65536	65534

19. AWS VPC Introduction

Amazon VPC

VPC = Virtual Private Cloud

Private network inside AWS.

20. Real-Time AWS Architecture

Example company network:

10.0.0.0/16

Why /16?

- Large environment
 - Multiple subnets
 - Future scalability
-

21. VPC Binary Understanding

VPC:

10.0.0.0/16

Binary:

00001010.00000000.00000000.00000000

Mask:

11111111.11111111.00000000.00000000

22. VPC Range Planning

Example:

Environment	CIDR
Entire VPC	10.0.0.0/16
Public	10.0.1.0/24
Application	10.0.2.0/24
Database	10.0.3.0/24

23. Subnet Deep Explanation

Subnet divides VPC.

Example:

10.0.1.0/24

Mask:

255.255.255.0

Hosts:

254 usable

24. AWS Reserved IPs

AWS reserves 5 IPs.

Example:

10.0.1.0/24

Reserved:

IP	Purpose
10.0.1.0	Network
10.0.1.1	Router
10.0.1.2	DNS
10.0.1.3	Reserved
10.0.1.255	Broadcast

25. Public Subnet

Has internet access.

Route:

0.0.0.0/0 → Internet Gateway

Used for:

- Bastion
 - Load Balancer
 - NGINX
-

26. Private Subnet

No direct internet.

Used for:

- Databases
 - Internal applications
-

27. Real-Time AWS Subnet Design

Production Example

Layer	CIDR
Public-AZ1	10.0.1.0/24

Layer	CIDR
Public-AZ2	10.0.2.0/24
App-AZ1	10.0.11.0/24
App-AZ2	10.0.12.0/24
DB-AZ1	10.0.21.0/24
DB-AZ2	10.0.22.0/24

28. Real-Time Server IP Assignment

Server	IP
NGINX	10.0.1.10
Tomcat-1	10.0.11.10
Tomcat-2	10.0.12.10
MySQL	10.0.21.10

29. Manual Private IP Selection in AWS

EC2 Launch:

```
Launch Instance
→ Network Settings
→ Advanced Network Configuration
→ Primary IP
→ Enter IP
```

Example:

```
10.0.11.10
```

30. Internet Flow in AWS

```
Internet
↓
Internet Gateway
↓
Public Subnet
↓
Load Balancer
↓
Private App Servers
↓
Database
```

31. NAT Gateway Flow

Private subnet internet access:

```
Private EC2
  ↓
NAT Gateway
  ↓
Internet Gateway
  ↓
Internet
```

32. Binary Subnetting Example

Suppose:

192.168.1.0/24

Need 4 subnets.

Borrow:

2 bits

New CIDR:

/26

Subnets

Subnet	Range
192.168.1.0/26	0-63
192.168.1.64/26	64-127
192.168.1.128/26	128-191
192.168.1.192/26	192-255

33. Why CIDR is Important in AWS

Used in:

- VPC
- Subnets
- Security Groups
- Route tables

- Kubernetes

34. Kubernetes Networking

Kubernetes needs large IP range.

Example:

10.0.32.0/20

IPs:

4096

35. Important Interview Questions

Why /16 VPC?

Scalability.

Why private subnet?

Security.

Why NAT Gateway?

Internet access for private servers.

Why multiple AZs?

High availability.

36. Best Practice Production Design

VPC: 10.0.0.0/16

Public:

10.0.1.0/24

10.0.2.0/24

Application:

10.0.11.0/24

10.0.12.0/24

Database:

10.0.21.0/24

10.0.22.0/24

37. Final Summary

Class A

- Huge networks

Class B

- Medium enterprise

Class C

- Small networks

CIDR

Modern subnetting method.

AWS Best Practice

- VPC → /16
- Subnet → /24